JAVA PROGRAMMING
Course Code – R1UC304C
SEM–III (Session 2025-26)

HEALTHCARE MANAGEMENT SYSTEM

TEAM MEMBERS:

HARSH ANAND (24SCSE1010085)

SATYAM KUMAR (24SCSE1010190)

SURAJ YADAV (24SCSE1570017)

Project Introduction

- • Web-based Healthcare Management System
- • Manages doctors, patients, appointments
- • Uses Java Servlet technology

Objectives of the Project

- • Implement Servlet-based web application
- • Perform CRUD operations using JDBC
- • Secure healthcare data
- • Session-based authentication
- • User-friendly interface

Technologies Used

- • Frontend: JSP, HTML, CSS
- • Backend: Java Servlets
- • Database: MySQL
- • Server: Apache Tomcat 9
- • IDE: NetBeans
- • JDK: 17+

System Architecture

- • MVC based architecture
- • View: JSP Pages
- • Controller: Servlets
- • Model: MySQL Database

- Flow: JSP → Servlet → DB → Servlet → JSP

Servlet Implementation

- • LoginServlet
- • LogoutServlet
- • DoctorServlet
- • PatientServlet
- • AppointmentServlet

- Each servlet extends HttpServlet

LoginServlet Working

- • Accepts username & password
- • Validates user using JDBC
- • Creates HttpSession
- • Redirects to dashboard
- • Handles invalid login

Session Management

- • Session created after login
- • Session attribute: username
- • Protected pages using session check
- • Logout destroys session

Doctor Management Module

- • Add doctor details
- • View doctor list
- • Data stored in MySQL
- • Handled using DoctorServlet

Patient Management Module

- • Register patient
- • View patient details
- • Data handled by PatientServlet

Database Design

- • user table
- • doctor table
- • patient table
- • appointments table

- Each table has primary key

Code Quality & Execution

- • Clean package structure
- • No SQL in JSP
- • Reusable DBConnection class
- • Proper exception handling
- • Efficient execution

Error Handling

- • Invalid login handling
- • Session validation
- • Try-catch blocks
- • User-friendly messages

Innovation / Extra Effort

- • Session-based security
- • Professional UI using CSS
- • Gradient buttons
- • Card-based layout
- • Scalable design

Project Flow

- Login → Dashboard
- → Add / View Doctor
- → Add / View Patient
- → Logout

Advantages

- • Secure
- • Easy data management
- • User-friendly
- • Scalable
- • Time efficient

Future Enhancements

- • Online appointment booking
- • Admin roles
- • Email notifications
- • Report generation

Conclusion

- • Successfully implemented Java Servlets
- • Secure session handling
- • Clean & modular code
- • Professional healthcare system

THANKS